

[1. Windows Event Logs Overview]

What are Windows Event Logs?

- Windows event logs are a forensic timeline of everything happening inside the Windows operating system.
- Every action triggers telemetry and generates an event:
 - User login → Event generated.
 - Process starts → Event generated.
 - Service starts → Event generated.
 - Policy change → Event generated.

Why They are Critical for SOC

- They provide definitive answers to crucial investigative questions: **WHO did WHAT, WHEN, FROM WHERE, and HOW.**
- Without logs, security investigations are impossible.

Where Logs are Stored

- **Directory Path:** C:\Windows\System32\winevt\Logs\
- **File Format:** .evtx

[2. Windows Logging Architecture]

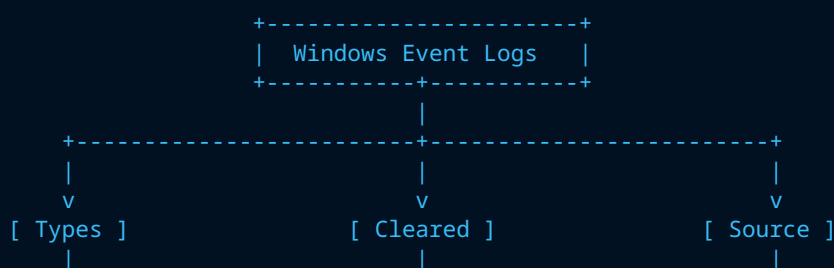
The Anatomy of a Windows Event Log

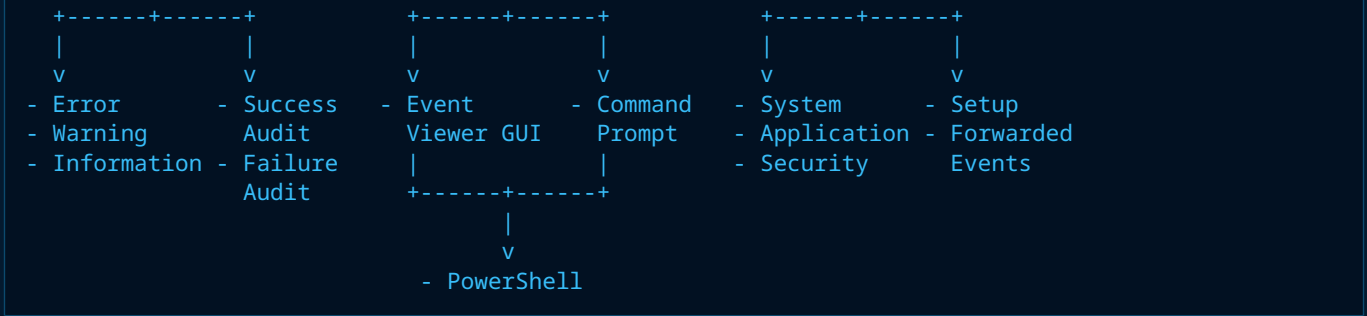
(Source: *hackthebox*)

An event log contains the following structural components:

- **Log name:** The name of the specific event log.
- **Source:** The software component or application that logged the event.
- **XML data:** All the structured log information formatted inside an XML layout.
- **Event ID:** A unique numerical identifier for the specific event type.
- **Task category:** A value or name explaining the purpose or use case of the event.
- **Computer:** The exact name of the computer host where the event took place.
- **Level:** The structural severity classification of the event.
- **Logged:** The precise date and time when the event was recorded.
- **Keywords:** Detailed category classifications used for sorting events.
- **OpCode:** Identifies the exact operation that the event reports.
- **User:** The specific user account that was logged on when the event occurred.

Windows Logging Architecture Mapping





Step-by-step Logging Flow

1. An action happens within the system (e.g., user login, process execution).
2. A dedicated **Event Provider** generates the corresponding event data.
3. The **Windows Event Log Service** collects the telemetry.
4. The event is permanently written to the corresponding .evtx log file.
5. The data is viewed by administrators in Event Viewer or parsed directly into a SIEM.

Key Logging Components

- **Event Provider:** Generates logs (e.g., Security provider, PowerShell logs).
- **Event Channel (Log Type):** Categories like Security, System, and Application.
- **Event Log Service:** The core operating system service that stores logs.

[3. Security Logs (Most Important for SOC)]

What are Security Logs?

Security logs are your primary investigation logs. They track:

- Authentication and authorization events.
- System privileges assignments.
- User account modification changes.

Crucial Logon Types Reference Table

Type	Meaning	SOC Use Case / Analysis
2	Interactive	Local keyboard/screen login on the machine.
3	Network	Network connection access (e.g., SMB share access).
7	Unlock	Desktop session reuse/unlocking a locked screen.
10	Remote	Remote Desktop Protocol (RDP) login.

SOC Rule: Analysts must map: **Logon Type + Source IP + Username = Attack Pattern.**

[4. Security Event Logs – Key Event IDs]

Logon & Authentication Events

- **4624 (Successful Logon):** Monitors user login activities. (Check for Type 2 Interactive, Type 10 RDP, or Type 3 Network).
- **4625 (Failed Logon):** Used to detect brute-force or credential stuffing attacks.
- **4634 (Logoff):** Standard user logoff event.
- **4647 (User Initiated Logoff):** User explicitly selected logoff.

- **4648 (Logon using Explicit Credentials):** Indicates possible credential abuse (e.g., RunAs, net use commands, or scheduled tasks).
- **4672 (Special Privileges Assigned):** Admin or highly privileged user logon (Very Important to track).

Account Management Events

- **4720:** New user account created.
- **4722:** User account enabled.
- **4723:** Password change attempt by the user.
- **4724:** Password reset initiated by an Administrator.
- **4725:** User account disabled.
- **4726:** User account deleted.

Group & Privilege Management

- **4728:** Member added to a global group.
- **4729:** Member removed from a global group.
- **4732:** Member added to a local security group.
- **4733:** Member removed from a local group.
- **4756:** Member added to a universal group.
- **4757:** Member removed from a universal group.
- **4673:** Sensitive privilege used.
- **4674:** An operation was performed on a privileged object.

CRITICAL FOCUS (HIGH ALERT): Pay extreme attention to any user added to the **Administrators** or **Domain Admins** groups.

Process Creation & Execution

- **4688 (Process Created): The Most Important Event!** Tracks new processes, parent processes, command lines, and user contexts.
- **4689 (Process Terminated):** Process termination event.

Domain Authentication (Kerberos / NTLM)

- **4768:** Kerberos TGT Request (Initial ticket-granting ticket requested).
- **4769:** Kerberos Service Ticket requested.
- **4771:** Kerberos Pre-authentication failed.
- **4776:** NTLM authentication attempt.
- **DETECTION USE:** Multiple Kerberos failures point to password spraying or brute force attacks. High NTLM usage shows legacy authentication or potential NTLM relay abuse.

Policy & Audit Changes

- **4719:** Audit policy was changed.
- **4739:** Domain policy was changed.

HIGH RISK: Threat actors frequently disable auditing configuration parameters to mask malicious footprints.

Object Access Events

- **4656:** A handle was requested to an object.
- **4663:** An attempt was made to access an object (files, registry, etc.).
- **4658:** A handle to an object was closed.

System & Security State Events

- **4608:** Windows started (System startup).
- **4609:** Windows shutdown.
- **4616:** System time was changed.
 - *Detection:* Unauthorized time changes can indicate log timeline manipulation.

Log Tampering (Critical Alert)

- **1102 (Security Log Cleared): IMMEDIATE INCIDENT RESPONSE REQUIRED.**

[5. Attack Detection Cheat Sheet for Security Logs]

- **Brute Force Attack:** Multiple 4625 (Failed Logon) events from the same IP address or targeting the same username.
- **Successful Attack:** A string of 4625 events immediately followed by a 4624 (Successful Logon) event.
- **Privilege Escalation:** Event 4672 running in tandem with Group membership change logs.
- **Malware Execution:** Event 4688 containing highly suspicious process paths or heavily encoded command lines.
- **Persistence:** Detection of a newly created account (4720), service installation (7045), or unauthorized scheduled tasks.
- **Log Tampering:** Event 1102 (Security Log Cleared) triggers a high priority investigation.

[6. System Logs]

What are System Logs?

System logs track operating system behaviors, driver states, hardware warnings, and system boot/power metrics.

Critical Windows System Logs Event IDs Reference

Category	Event ID	Description / Telemetry Meaning
Startup & Shutdown	6005	Event Log Service Started (System boot/startup).
	6006	Event Log Service Stopped (Normal clean shutdown).
	6008	Unexpected Shutdown (System crashed or lost power).
	6013	System Uptime (Shows total execution duration).
Power & Reboot	41	Kernel-Power (Rebooted without clean shutdown / BSOD).
	1074	Planned Shutdown/Restart initiated by user or process.
	109	Kernel Power Critical Error.
Drivers & Hardware	219	Driver Failed to Load during system startup.
	225	Device Not Migrated properly.
Disk & Storage	11	Disk Controller detected an error.
	15	The storage device is not ready for use.
	7	A bad block was detected on the physical disk.
	51	Disk Warning (An I/O operation was retried or failed).

Category	Event ID	Description / Telemetry Meaning
	55 / 98	NTFS file system or volume structure is corrupted.
Service Control	7000	A Windows Service failed to start.
	7001	Service Dependency Failed to initialize.
	7009 / 7011	Service startup or response timeout.
	7034	A running background service crashed unexpectedly.
	7036	Service State Changed (Running / Stopped / Paused).
	7045	New Service Installed (Crucial for finding malware).
Network & Comms	4201 / 4202	Network Adapter Connected or Disconnected.
	1014	DNS Client Events (DNS name resolution failed).
Time & Integrity	1	The local system time was synchronized.
	36	Time Service encountered an error.
Crash Reporting	1000	Application Error (An application crashed).
	1001	BugCheck (A Blue Screen of Death - BSOD occurred).

System Health Indicators & SOC Shortcuts

- **System Crash/Reboot:** Combination of 41 + 6008 + 1001.
- **Disk Failure:** Triggered by IDs 7, 51, 55, or 98.
- **Service Failure:** Marked by IDs 7000 or 7034.
- **Driver Issue:** Identifiable via 219 or 225.
- **Network Issue:** Indicated by ID 1014 (DNS failures).
- **SOC Use:** Detect anomalous system crashes that could indicate exploit attempts.

[7. Application Logs]

What They Track

- Application crashes, application hangs, software execution errors, and general software activity.

SOC Relevance

- Used to detect application exploit attempts and identify malicious or unapproved software behaviors. For example: If Microsoft Word crashes (1000), it could indicate a failed buffer overflow exploit attempt.

Application Event IDs At A Glance Table

Event ID	Description / Log Action
1000	Application Error (Crashed or stopped working unexpectedly).
1001 / 1002	Application Hang / Application Unresponsive over a timeframe.
1003 / 1004	Application Restart / Application Started successfully.
1005 / 1006	Application Stopped successfully / Application Crash event.

Event ID	Description / Log Action
1007	Application Fault occurred.
1026 / 1027	.NET Framework application runtime error or warning.
1028 / 1029	.NET Framework runtime informational or debugging events.
1030 / 1031	Application Installed / Application Uninstalled.
1032 / 1033	Application Updated / Configuration profile was changed.
1034	Error detected in application configuration.
1040 / 1041	Application Access Denied / Invalid Credentials used by the app.
1042 / 1043	Application Login Failure / Permissions configuration changed.
1044	A software security-related error occurred.
1050 / 1051	Performance Warning / Critical Performance Error.
1052 / 1053	Insufficient resources available / Critical resource unavailable.
1054	The application operation timed out.
1060 / 1061	Service Communication Failure / Error talking to external system.
1062 / 1063	Message Queue Error / Application failed to send email.
1064	Application API call failed.
1070 / 1071	Application Audit Success / Audit Failure.
1072 / 1073	Log entry written successfully / Failed to write log entry.

[8. Windows Process Architecture]

Process Definition

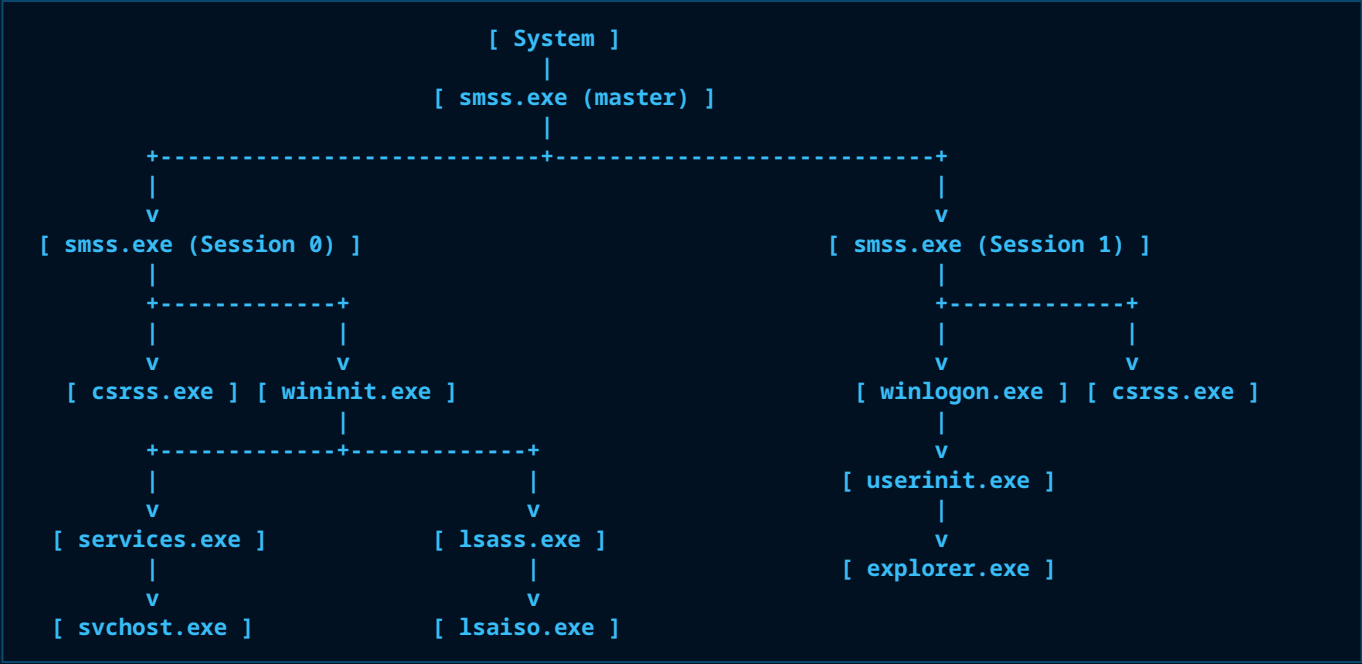
A process is a live, running instance of a program executing in memory, containing its own:

- Isolated virtual memory space.
- Execution threads.
- System handles and resources.
- Explicit security context.
- *Simple View*: A programmatic binary file sitting on a disk (e.g., chrome.exe) when executed becomes a process in memory assigned a unique Process ID (PID).

Key Components of a Process

1. **PID (Process ID):** A unique numerical identifier for the running instance (e.g., PID: 4321).
2. **Virtual Memory:** Each process gets isolated memory allocation. (SOC Note: Target for process injection attacks).
3. **Threads:** Core units of execution inside a process. One process can contain multiple threads.
4. **Handles:** Structural references to open system resources, such as files, registry keys, or network sockets.
5. **Security Context:** Defines the precise user privilege tier the process runs under (e.g., SYSTEM, Administrator, or Standard User).

Windows Core Processes Tree



(Credit: @halfcircassian)

The Process Life Cycle

- **Waiting State:** A process shifts to a waiting state when it cannot run because it is waiting for user/network input, waiting for a signal, or restricted by the kernel to ensure fairness.

Execution Sequence Steps

1. Execution is triggered by a user action or system script.
2. The Win32 API is called.
3. The OS Kernel creates a new process object.
4. Dedicated virtual memory is allocated.
5. Execution threads are created.
6. The process begins executing code.

Parent-Child Relationship

- Every process is created by another process.
 - **Normal Example:** explorer.exe (Desktop interface) → spawns → chrome.exe.
 - **SOC Insight:** If the parent-child combination is non-standard, it is highly suspicious. For example: winword.exe → spawning → powershell.exe.

[9. Critical Windows System Processes Table]

Process Name	Process File	Typical PID Range	OS Importance & Description	Impact if Terminated
System	ntoskrnl.exe	4 - 500	CRITICAL: The Windows NT OS Kernel. Manages hardware, memory, resources.	Immediate system crash (BSOD).
smss.exe	smss.exe	500 - 600	CRITICAL: Session Manager Subsystem. Starts user sessions and core system processes.	System will fail to start or shut down.

Process Name	Process File	Typical PID Range	OS Importance & Description	Impact if Terminated
csrss.exe	csrss.exe	600 - 900	CRITICAL: Client/Server Runtime Subsystem. Handles Win32 subsystem/console windows.	Applications and system UI stop working.
wininit.exe	wininit.exe	900 - 1000	CRITICAL: Windows initialization process. Spawns services and userinit.exe.	System becomes completely unusable.
services.exe	services.exe	1000 - 1500	CRITICAL: Service Control Manager. Manages background Windows services.	Services stop running; system instability.
lsass.exe	lsass.exe	600 - 1200	CRITICAL: Local Security Authority Subsystem. Handles authentication and user logons.	User logons fail; security functions stop.
svchost.exe	svchost.exe	1000 - 6000+	CRITICAL: Generic host process for services. Multiple instances run simultaneously.	All dependent background services stop.
explorer.exe	explorer.exe	2000 - 4000	HIGH: Windows Explorer. Manages desktop GUI, taskbar, Start menu, and files.	Desktop GUI and taskbar disappear.
spoolsv.exe	spoolsv.exe	1500 - 3000	HIGH: Print Spooler. Manages print jobs and hardware printer comms.	Printing operations fail completely.
taskhostw.exe	taskhostw.exe	2000 - 5000	MEDIUM: Hosts tasks scheduled by users or the system via Task Scheduler.	Scheduled tasks may fail to execute.
dwm.exe	dwm.exe	1500 - 3000	MEDIUM: Desktop Window Manager. Renders OS visual effects and window structures.	Affects overall UI performance and visuals.
MsMpEng.exe	MsMpEng.exe	2000 - 5000	LOW: Microsoft Defender Antivirus engine process.	Real-time system protection disables.
wuaclt.exe	wuaclt.exe	2000 - 4000	LOW: Windows Update Client. Checks and installs system security updates.	Windows Updates stop working.

WARNING: Never terminate critical system processes as it induces severe OS instability or immediate system failure. Verify their path (should be in System32), check for a valid Microsoft Corporation digital signature, and analyze behavior before taking action.

[10. Process-Based Attack Techniques]

Process Classification

1. **User Processes:** Active user software applications.
2. **System Processes:** Core operating system-level engines.
3. **Background Processes:** Silent background system services.

Common Process Exploitation Vectors

- **Process Injection:** Code is injected directly into a separate, clean process (e.g., malware injecting code into explorer.exe). Look for suspicious child processes or memory allocation anomalies.
- **DLL Injection:** Forcing a process to load a malicious dynamic-link library file.
- **Process Hollowing:** A legitimate binary (e.g., notepad.exe) is started in a suspended state, its legitimate memory space is unmapped, and it is replaced with malicious payload code.
- **LOLBins (Living Off the Land Binaries):**** Utilizing completely legitimate, pre-installed administrative operating system tools like powershell.exe, cmd.exe, or rundll32.exe to execute attacks. A common SOC indicator is powershell.exe running with heavily encoded or hidden parameters.